```
100 'LOW-PASS WINDOWED-SINC FILTER
110 'This program filters 5000 samples with a 101 point windowed-sinc filter,
120 'resulting in 4900 samples of filtered data.
130 '
140 DIM X[4999]                    'X[ ] holds the input signal
150 DIM Y[4999]                    'Y[ ] holds the output signal
160 DIM H[100]                     'H[ ] holds the filter kernel
170 '
180 PI = 3.14159265
190 FC = .14                       'Set the cutoff frequency (between 0 and 0.5)
200 M% = 100                       'Set filter length (101 points)
210 '
220 GOSUB XXXX                     'Mythical subroutine to load X[ ]
230 '
240 '                              'Calculate the low-pass filter kernel via Eq. 16-4
250 FOR I% = 0 TO 100
260   IF (I%-M%/2) = 0 THEN H[I%] = 2*PI*FC
270   IF (I%-M%/2) <> 0  THEN H[I%] = SIN(2*PI*FC * (I%-M%/2)) / (I%-M%/2)
280   H[I%] = H[I%] * (0.54 - 0.46*COS(2*PI*I%/M%) )
290 NEXT I%
300 '
310 SUM = 0                        'Normalize the low-pass filter kernel for
320 FOR I% = 0 TO 100              'unity gain at DC
330   SUM = SUM + H[I%]
340 NEXT I%
350 '
360 FOR I% = 0 TO 100
370   H[I%] = H[I%] / SUM
380 NEXT I%
390 '
400 FOR J% = 100 TO 4999           'Convolve the input signal & filter kernel
410   Y[J%] = 0
420   FOR I% = 0 TO 100
430     Y[J%] = Y[J%] + X[J%-I%] * H[I%]
440   NEXT I%
450 NEXT J%
460 '
470 END
```

TABLE 16-1

a stopband attenuation of at least -120dB (one part in one-million for those that refuse to learn decibels). As previously shown, the Blackman window only provides -74dB (one part in five-thousand). Fortunately, greater stopband attenuation is easy to obtain. The input signal can be filtered using a conventional windowed-sinc filter kernel, providing an intermediate signal. The intermediate signal can then be passed through the filter a second time, further increasing the stopband attenuation to -148dB (1 part in 30 million, wow!). It is also possible to combine the two stages into a single filter. The kernel of the combined filter is equal to the *convolution* of the filter kernels of the two stages. This also means that convolving any filter kernel *with itself* results in a filter kernel with a much improved stopband attenuation. The price you pay is a longer filter kernel and a slower roll-off. Figure 16-7a shows the frequency response of a 201 point low-pass filter, formed by convolving a 101 point Blackman windowed-sinc with itself. Amazing performance! (If you really need more than -100dB of stopband attenuation, you should use double precision. Single precision